

Hall Ticket No. -

Code: R122131/1

Stanley College of Engineering & Technology for Women (A)

B.E (IT) - III Semester (Main) Examinations-January-2023

Probability & Statistics

Time: 03Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
iii. Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

1. a) A Random Variable X has the following probability distribution

X	1	2	3	4
P	1/10	2/10	3/10	4/10

Then find (i) $E(X)$ (ii) $V(X)$ [2M CO1 BTL1]

- b) Determine the B.D for which mean 4 and variance 3. [2M CO2 BTL2]

- c) For an F-distribution determine the following

(i) $F_{0.05}$ With $v_1 = 7$ and $v_2 = 15$ (ii) $F_{0.01}$ with $v_1 = 24$ and $v_2 = 19$. [2M CO3 BTL2]

- d) Solve $\frac{dy}{dx} = x + y$, $y(1) = 0$ numerically using Taylor's series method up to $x = 1.1$

with $h = 0.1$ [2M CO4 BTL2]

- e) Define Linearly Independent and dependent vectors in Vector spaces. [2M CO5 BTL2]

- f) Find the singular values of the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ [2M CO5 BTL2]

Part-B

4X12=48M

2. a) In a bolt factory machines A, B, C manufacture 20%, 30%, 50% of the total of their output and 6%, 3%, 2% are defective. A bolt is drawn at random and found to be defective. What is the probability that it is manufactured by machines A, B and C? [6M CO1 BTL2]

- b) If probability density function $f(x) = \begin{cases} kx^3, & \text{if } 0 \leq x \leq 3 \\ 0, & \text{else where} \end{cases}$

Find the probability between $x = \frac{1}{2}$ and $\frac{3}{2}$.

[6M CO1 BTL2]

3. a) Calculate the first four moments of the following distribution about the mean

x	0	1	2	3	4	5	6	7	8
f	1	8	28	56	70	56	28	8	1

And also calculate β_1, β_2

[8M CO2 BTL3]

- b) For the discrete probability distribution

X	0	1	2	3	4	5	6
F	0	2K	2K	3K	K^2	$2K^2$	$7K^2 + K$

Find (i) K (ii) Mean (iii) Variance

[4M CO2 BTL3]

4. a) Find the equation of the least fitting line
- $y = ax + b$
- for the following data

x	5	10	15	20	25
y	16	19	23	26	30

[6M CO3 BTL3]

- b) Find the Regression Co-efficient from the following data.

X	68	64	75	50	64	80	75	40	55	64
Y	62	58	68	45	81	60	68	48	50	70

[6M CO3BTL4]

5. a) Using Runge - Kutta method of fourth order find
- $y(0.1)$
- and
- $y(0.2)$
- given that

$$\frac{dy}{dx} = 1 + xy, y(0) = 2.$$

[12M CO4 BTL4]

6. a) Prove that the set
- $S = \{(1, -1, 0), (1, 1, 0), (1, 1, 1)\}$
- is a basis of
- $\mathbb{R}^3$

[6M CO5 BTL5]

- b) Find the singular values of the matrix
- $A = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{pmatrix}$

[6M, CO5, BTL5]

7. a) Fit a second degree parabola
- $y = ax^2 + bx + c$
- to the following data

x	0	1	2	3	4
y	1	5	10	22	38

[6M, CO3, BTL3]

- b) Given,
- $\frac{dy}{dx} = x^3 + y, y(0) = 1$
- , compute
- $y(0.2)$
- by Euler's Method taking
- $h = 0.01$

[6M CO4 BTL 4]

Stanley College of Engineering & Technology for Women

B.E (IT) III Semester (Main) Examinations-January-2023

Discrete Mathematics

Time: 03Hours

Max. Marks-60

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Part-A

6X2=12M

1. a. Define an Equivalence Relation with an example? [2M CO1 BTL1]
- b. State Pigeon hole Principle? [2M CO2 BTL2]
- c. Construct the Truth table of $P \wedge (P \rightarrow Q)$ [2M CO3 BTL3]
- d. Let $G = (1, -1)$. Verify that $\langle G, * \rangle$ is a Group. [2M CO4 BTL4]
- e. Write about Planar graphs with an example? [2M CO5 BTL2]
- f. Differentiate Path and Cycle of a graph? [2M CO6 BTL2]

Part-B

4X12=48M

2. a) Let $A = \{1, 2, 3, 4, 6, 12\}$. On A, define the relation R by aRb if and only if a divides b. Prove that R is a Partial Ordering Relation on A. Draw the Hasse Diagram for this relation. [6M CO1 BTL5]
- b) Consider $A = \{1, 2, 3, 4\}$ and f and g be functions from A to A given by $f = \{(1, 4), (2, 1), (3, 2), (4, 3)\}$ and $g = \{(1, 2), (2, 3), (3, 4), (4, 1)\}$. Prove that f and g are inverse of each other. [6M CO1 BTL5]
3. a) From a group of 10 professors, how many ways can a committee of 5 members can be formed, so that atleast one professor A and professor B will be included? [6M CO2 BTL2]
- b) Suppose that 200 faculty members can speak French and 50 can speak Russian, while only 20 can speak both French and Russian. How many faculty members can speak either French or Russian? [6M CO BTL2]
4. a) Show that $(P \rightarrow Q) \Leftrightarrow (\neg Q \rightarrow \neg P)$ through Truth Table Construction. [6M CO3 BTL2]
- b) List out the Connectives used in Propositional Logic. Explain? [6M CO3 BTL2]

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5. a) Let G be the set of all non-zero real numbers and let $a * b = (\frac{1}{2})ab$. Show that $\langle G, * \rangle$ is an abelian group. [6M CO4 BTL6]

b) How the Disjunctive Normal Form is different from Conjunctive Normal Form. Explain? [6M CO4 BTL5]

6. a) Explain Isomorphism of Graphs with an Example? [6M CO5 BTL6]

b) Explain Proper Coloring and Chromatic Number of a graph with an example? [6M CO BTL6]

7. a) Explain Euclidean Algorithm with an example? [6M CO1 BTL2]

b) Find the coefficient of x^{15} in the expansion of $(x - x^2)^{10}$. [6M CO2 BTL2]

Stanley College of Engineering & Technology for Women (A)

B.E (IT) III Semester (Main) Examinations-January-2023

Digital Electronics & Logic Design

Time: 03 Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
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Part-A

6X2=12M

1. a. Construct XOR gate using only NAND Gates [2M CO1 BTL3]
- b. What do you mean by priority encoder? [2M CO2 BTL1]
- c. Define LUT and draw 2 – input LUT. [2M CO3 BTL1]
- d. What are direct inputs in a flip-flop and why they are used? [2M CO4 BTL2]
- e. Compare Moore and Mealy machine? [2M CO5 BTL2]
- f. State De Morgan's Theorem. [2M CO1 BTL1]

Part-B

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2. a) Determine the set of prime implicants and obtain all the minimal exp function $F(w,x,y,z)=\Sigma(0,1,2,5,7,8,9,10,13,15)$
- b) Simplify the Boolean expression $(x+y)(x'+y')$ to minimum number of literals and implement the logic using universal gates. [6M CO1 BTL4]
3. a) Explain the 3x8 line Decoder with Truth-table and circuit diagram. [6M CO2 BTL2]
- b) Construct a Full Subtractor using 3x8 line decoder. [6M CO2 BTL3]
4. a) Explain the architecture of CPLD. [6M CO3 BTL1]
- b) Build the PAL programming table for the Boolean functions $w=A+BC+BD$, $x=B'C+B'D+BC'D'$, $y=CD+C'D'$, $z=D'$ [6M CO3 BTL6]
5. a) What is a master-slave flip-flop? Explain with block diagram and logic diagram. [6M CO4 BTL1]
- b) Design a T flip-flop using JK flip-flop. Use k-maps for the design. [6M CO4 BTL3]
6. a) List the basic types of shift registers in terms of data movement. [6M CO4 BTL1]
- b) Design a mod 6 ripple counter using T flip-flops [6M CO4 BTL3]
7. a) Convert the following Mealy machine to Moore Machine [6M CO5 BTL4]

PS	NS,Z	
	x=0	x=1
A	C,0	B,1
B	A,1	D,0
C	B,1	A,1
D	D,1	C,0

- b) Explain about the basic elements of ASM charts

[6M CO5 BTL2]

Hall Ticket No. -

Code: R122135

Stanley College of Engineering & Technology for Women (A)

B.E (IT) III Semester (Main) Examinations-January-2023

Database Management Systems

Time: 03 Hours

Max. Marks-60

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
iii. Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

1. a. Define Instances and schemas of database?
- b. List the aggregate functions supported by SQL?
- c. Demonstrate functional dependency? Give an example?
- d. Define a Transaction?
- e. Discuss about data on External storage?
- f. What is a Sparse index?

[2M CO4 BTL1]

[2M CO5 BTL1]

[2M CO4 BTL1]

Part-B

4X12=48M

2. a) Explain about Database users and Administrators? [4M CO1 BTL1]
b) Explain about Database Architecture [8M CO1 BTL1]
3. a) Define trigger and explain its three parts? Compare row level and statement level triggers? [6M CO3 BTL3]
b) Explain Set operations of Relational Algebra with examples? [6M CO2 BTL3]
4. a) State and explain various features of E-R Models [6M CO3 BTL2]
b) Explain 3NF. What are the steps to be followed to convert a relation in 2NF to 3NF? [6M CO4 BTL2]
5. a) What is an index? Explain sparse index with examples. [6M CO3 BTL3]
b) How to test serializability of a schedule? Explain with an example. [6M CO5 BTL2]
6. a) Explain in detail about the two-phase locking protocol [6M CO5 BTL3]
b) Explain in detail about Timestamp-Based Protocols? [6M CO5 BTL3]
7. a) Explain in detail about Lock-Based Protocols [7M CO5 BTL3]
b) Explain about Buffer Management [5M CO4 BTL3]

Stanley College of Engineering & Technology for Women (A)**B.E (IT) III Semester (Main) Examinations-January-2023****OOPS Using JAVA****Time: 03 Hours****Max. Marks-60**

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
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Part-A**6X2=12M**

1. a. Illustrate the concept of type conversion and casting. [2M CO1 BTL3]
b. Explain the usage of java packages. [2M CO2 BTL2]
c. Describe how we can create a thread [2M CO3 BTL2]
d. List any two classes from java.lang package with their importance. [2M CO4 BTL1]
e. Explain any two adapter classes along with their usage. [2M CO5 BTL2]
f. Define String Buffer Class. [2M CO1 BTL1]

Part-B**4X12=48M**

2. a) Describe the benefits and applications of object-oriented programming. [6M CO1 BTL2]
b) Define class, method and object? Show the syntax to define these in java. [6M CO1 BTL1]
3. a) State and explain various types of constructors. [6M CO1 BTL2]
b) What is Package? Demonstrate creating a package in java with example. [6M CO2 BTL3]
4. a) Explain the concept of serialization in I/O Streams. [6M CO4 BTL2]
b) Illustrate the member access mechanism in inheritance with an example. [6M CO2 BTL4]
5. a) Define Daemon threads. Explain with an example. [6M CO3 BTL1]
b) Demonstrate the exception handling mechanism of Java. [6M CO3 BTL3]
6. a) Explain the Java collection framework. [6M CO4 BTL2]
b) Describe any two from the following with a simple example.
i) Autoboxing [3M CO5 BTL2]
ii) Java Annotations [3M CO5 BTL2]
iii) String Tokenizer [3M CO4 BTL2]
7. a) Explain the difference between AWT and Swing. [6 M CO5 BTL2]
b) Write a java program to implement Border Layout. [6M CO5 BTL3]

Stanley College of Engineering and Technology for Women(A)**B.E(EEE,ECE & IT) II Semester (Backlog) Examinations January-2023****Indian Constitution**

Time: 3 hours

Max.Marks:60

PART-A**5 X 2=10M****Note: Answer all questions(Compulsory)**

- | | |
|---|------------|
| 1 Who are the members of Drafting Committee? | 2 M CO1 L1 |
| 2 What are Panchayati Raj Institutions?Name them. | 2 M CO2 L2 |
| 3 Explain about Article-19. | 2 M CO3 L2 |
| 4 Why is NITI Aayog Necessary? | 2 M CO4 L1 |
| 5 What is the Composition of National Commission for Women? | 2 M CO5 L3 |

PART-B**5 X 10= 50M****Note: Answer all questions**

- | | |
|---|-------------|
| 6 a. Discuss the Significance of Indian Constitution. | 5 M CO1 L4 |
| b. What are the Features of 1935 Act? | 5 M CO1 L1 |
| OR | |
| c. Explain the Philosophical Foundations to the Indian Constitution. | 10 M CO1 L3 |
| 7 a. Examine the Powers and Functions of Prime Minister. | 5 M CO2 L4 |
| b. Explain about Emergency powers of President of India. | 5 M CO2 L2 |
| OR | |
| c. Write about Evolution of Local Government in India. | 10 M CO2 L4 |
| 8 a. Explain the Importance of Fundamental duties. | 5 M CO3 L3 |
| b. Briefly explain about DPSP. | 5 M CO3 L4 |
| OR | |
| c. Explain the features of Fundamental rights. | 5 M CO3 L2 |
| d. Explain the Five Writs in Indian Constitution. | 5 M CO1 L1 |
| 9 a. Explain about Composition of Inter State Council. | 5 M CO4 L3 |
| b. Write an essay on Role and Functions of Finance commission of India. | 5 M CO4 L4 |
| OR | |
| c. Explain about the functions of NITI Aayog. | 5 M CO4 L2 |
| d. Explain about Administrative relations of Union-State. | 5 M CO4 L4 |
| 10 a. What are the Powers and Functions of NHRC? | 10 M CO5 L4 |
| OR | |
| c. What are the Powers and Functions of National Commission for Women? | 10 M CO5 L1 |

Hall Ticket No-

Code:22315/1

Stanley College of Engineering and Technology for Women(A)

B.E (ECE, EEE & IT) II Semester (Backlog) Examinations January-2023

Essence of Indian Traditional Knowledge

Time: 3 hours

Max.Marks:60

PART-A

Note:Answer all questions (Compulsory)

5 X 2=10M

- 1 Define Civilization. 2M L1
- 2 What are Brahmanas and Aryankas? 2M L1
- 3 List the Heterodox schools of Indian Philosophy. 2M L1
- 4 Write a short note on Sama Veda. 2M L1
- 5 What were the subjects of study in Ancient India? 2M L1

PART-B

Note: Answer all questions

5 X 10= 50M

- 6 a. Discuss the general characteristics of Culture. 5M L2
b. Explain the uniqueness of Indian Culture. 5M L2
OR
c. Write about ancient India. 5M L1
d. Write an essay on the influence of the English in Modern India. 5M L1
- 7 a. Summarize the developments of South Indian Languages and Literature. 5M L2
b. Explain about the Vedas and Upanishads. 5M L2
OR
c. Discuss the classic Sanskrit literature. 5M L2
d. Discuss the literature of North Indian Languages. 5M L2
- 8 a. Examine the salient features of orthodox schools of Indian Philosophy. 5M L2
b. What is the important role played by the Bhakti and Sufi religious reform movements in India? 5M L2
OR
c. What is Sunyavada of Buddhism?Explain. 5M L2
d. Discuss the Religious reform movements in Modern India. 5M L2
- 9 a. Write an essay on "Ancient Indian Architecture". 5M L2
b. Explain elaborately about the Fine Arts of India . 5M L2
OR
c. Discuss about Indian classical dance forms. 5M L2
d. Justify the statement "Science and technology can solve all the world's problems, and historically it has been shown to make the world better and better." 5M L3
- 10 a. Write about the education system in Medieval India. 5M L1
b. List any five contributions made by Ancient Indian scientists to the world of Science. 5M L1
OR
c. Explain the Gurukula Education system. 5M L2
d. Discuss the role of Education in social change. 5M L1

Hall Ticket No-

Code: R222823

Stanley College of Engineering and Technology for Women(A)

B.E (ECE & EEE) II Semester (Backlog) Examinations January-2023

Engineering Physics

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- 1 A plane transmission grating having 6000 lines per cm is used in normal incidence mode. Calculate the angular separation between the two spectral lines of wavelengths 5890 Å and 5896 Å in second order. 2 M
- 2 Explain the phenomenon of total internal reflection. 2 M
- 3 Define a) Space lattice and b) Unit cell. 2 M
- 4 Calculate the wavelength of an electron accelerated to a potential of 900 V. 2 M
- 5 What is Lenz's law? 2 M

PART-B

Note: Answer all questions

5 X 10= 50M

- 6 a. Explain the Young's double slit experiment to produce the interference pattern and obtain the conditions for maxima and minima. 5 M
- b. Discuss intensity expression for N slit diffraction grating. 5 M
- OR
- c. Analyze interference of light by division of amplitude and division of wave front with examples. 5 M
- d. Classify Fresnel's and Fraunhofer's diffraction patterns. 5 M
- 7 a. Explain the construction and working of semiconductor laser. 5 M
- b. Explain the types of optical fibers basing on refractive index profiles. 5 M
- OR
- c. Analyze the basic components required for producing laser 5 M
- d. Derive an expression for Numerical aperture and acceptance angle of an optical fiber. 5 M
- 8 a. Deduce the equation for the interplanar distance for cubic crystal. 5 M
- b. Classify the point defects in a crystal. 5 M
- OR
- c. Formulate the expression for the concentration of Schottky defects in ionic crystals. 5 M
- d. What are Miller indices? How to obtain them for a set of lattice planes? 5 M
- 9 a. Formulate the Schrodinger's time independent wave equation. 5 M
- b. Explain polarization in dielectric materials. 5 M
- OR
- c. Summarize the properties of wave function Ψ and explain its significance. 5 M
- d. Determine an expression for electric field due to continuous charge distribution. 5 M
- 10 a. Explain the concept of displacement current in dielectrics. 5 M
- b. Explain Faraday's law and Lenz's law. 5 M
- OR
- c. Obtain Poynting theorem. 5 M
- d. Build the Maxwell's equations for static fields. 5 M

Code: S222824/1

- 8 a Outline any five experimentally observed properties of superconductors. 7M CO3 L6
- b Explain Josephson Effect in superconductivity. 3M CO3 L2

OR

- c Explain general properties of superconducting materials 7M CO3 L2
- d Analyze de-Broglie's concept of matter waves. 3M CO3 L4

- 9 a What are p-type and n-type materials? 5M CO4 L3
- b Summarize assumptions and failures of classical free electron theory. 5M CO4 L4

OR

- c Explain the concept of energy band formation using Kronig-Penney model. 6M CO4 L3
- d Explain the working of Thermistor. 4M CO4 L2
- 10 a Explain thermal evaporation technique to prepare thin films. 6M CO5 L2
- b Discuss any three properties of materials which get affected at reduced size 4M CO5 L2

OR

- c Explain Ball milling method to prepare Nano materials. 6M CO5 L2
- d Distinguish bulk, thin and Nano materials. 4M CO5 L2

Hall Ticket No :

Code :S222821

Stanley College of Engineering and Technology for Women (A)

B.E(ECE,EEE & IT) II Semester (Backlog) Examinations January-2023

English

Time : 3 hours

Max.Marks : 60

PART-A

Note : Answer all questions (Compulsory)

5 X 2=10M

2 M BL1 CO1

1 Select the appropriate answer and fill in the blanks. (1/2m x 4 = 2 M)

- a. She does all her work _____ great care. (with/ by)
- b. I saw myself _____ the mirror. (in/on)
- c. The necklace was _____ present from her parents. (a/an/the)
- d. They were going to _____ concert. (Article) (a/ an/ the)

2 Use the appropriate tense forms of the given verbs and fill in the blanks.(1/2 m x4 = 2M) 2 M BL1 CO2

- 1. How have you been _____ ? (do)
- 2. He was badly wounded and _____ (bleed) profusely.
- 3. I usually _____ (teach) Chemistry, but I _____ (teach) Biology this year.
- 4. She _____ (finish) her breakfast and went to school.

3 Select the correct answer. (1/2m x 2 = 1 M)

2 M BL4 CO3

- a. A study of coins. _____ (numismatics/balistics)
- b. Handwriting which is difficult to read. _____ (legible/illegible)
Rewrite as directed.(1/2m x 2 = 1 M)
- c. Lavanya said, " We painted the house yesterday (Indirect speech)."
- d. She said, "I gave the key to my roommate (Indirect Speech)."

4 Select the appropriate word and fill in the blanks. (1/2mx 2 = 1M)

2 M BL1 CO4

- a. Let us _____ to the auditorium. (precede, proceed)
- b. The bus _____ (fare/fair) increased drastically.

Change the voice. (1/2m x 2 = 1M)

- c. The students have paid the fees.
- d. She was congratulated by her friends.

5 Rewrite the sentences correctly (1/2m x 4 = 2 M)

2 M BL3 CO5

- a. I will look forward to see you soon.
- b. Please don't repeat the mistake again.
- c. The water is very hot to drink.
- d. Harsha don't know Hindi.

PART-B

Note : Answer all questions

5 X 10= 50M

6 a. Summarize " On the Conduct of Life" by William Hazlitt briefly.

5 M BL2 CO1

b. Use the following hints to write a paragraph, give it a title and a moral.

5 M BL3 CO1

Hints : The King of Scotland was defeated six times-----hid himself in a cave saw a spider trying to climb up to the roof of the cave the spider fell six times and then succeeded- the king was encouraged - tried again to defeat his enemies and succeeded.

OR

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- c. Summarize the young boy's attitude towards his new school as explained in William Hazlitt's "On the Conduct of Life." 5 M BL2 CO1
- d. Examine the structure of writing an effective paragraph. 5 M BL3 CO1
- 7 a. Outline the central idea of the poem "If." 5 M BL2 CO2
- b. Describe compounding, clipping and blending words with examples. 5 M BL1 CO2
- OR
- c. Outline the poet's views on honesty and fortitude of character from the poem "If." 5 M BL1 CO2
- d. Describe briefly "Recent floods due to continuous rains." 5 M BL2 CO2
- 8 a. Discuss the dream that Martin Luther King envisions for America. 5 M BL2 CO3
- b. Describe your childhood memories. 5 M BL2 CO3
- OR
- c. Discuss "I have a Dream" as a plea for freedom and equality. 5 M BL2 CO3
- d. Describe your views on "Is country life better than city life." 5 M BL2 CO3
- 9 a. Examine the conflict of the Poem "The Road not Taken." 5 M BL3 CO4
- b. Justify the topic "Should the minimum voting age be raised/lowered/kept the same?" 5 M BL5 CO4
- OR
- c. Examine the spirit of regret and longing in the poem "The Road not Taken." 5 M BL3 CO4
- d. Justify the topic "Is online learning effective?" 5 M BL5 CO4
- 10 a. Explain the important features of a blog. 5 M BL2 CO5
- b. Create a blog on the topic: "Writing captions on instagram." 5 M BL6 CO5
- OR
- c. Explain the process of writing "Tweets" and "Instagrams." 5 M BL2 CO5
- d. Create a tweet in response to the proposal of the "New Labour Code.i.e. 4- day work per week, 12 working hours every working day in the week." 5 M BL6 CO5

Hall Ticket No-

Code S222825

Stanley College of Engineering and Technology for Women (A)

B.E (CSE/EEE/ECE/IT/CME/AI&DS) II Semester (Backlog) Examinations January-2023

Mathematics--II

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5X2=10M

- 1 Solve $(x^2 + y^2) dx + 2xy dy = 0$ 2 M L3
- 2 Find the C.F. of $(D^2 + D + 1)y = 0$ 2 M L2
- 3 Define Beta function and Gamma function 2 M L1
- 4 Find the Eigen Values of the Matrix $A = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$ 2 M L3
- 5 Find $L\{t^2 \sin at\}$ 2 M L3

PART-B

Note: Answer all questions

5X10=50M

- 6 a. Solve $\frac{dy}{dx} + y \cos x = y^3 \sin 2x$ 5 M L3
- b. Find the Orthogonal trajectories of $\frac{x^2}{a^2 + \alpha} + \frac{y^2}{b^2 + \alpha} = 1$, α being the parameter. 5 M L3
- OR
- c. Solve $(1 + e^{xy}) dx + e^{x/y} (1 - \frac{x}{y}) dy = 0$ 5 M L3
- d. Solve $p = \sin(y - xp)$. Also find its singular solutions. 5 M L3
- 7 a. Solve $(D^2 + 4)y = e^x + \sin 2x + \cos 2x$ 5 M L3
- b. Solve $y'' + y = \sec x$ by the Method of Variation of parameters. 5 M L3
- OR
- c. Solve $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + 2y = x \log x$ 5 M L3
- d. Solve $(D^2 - 4D + 3)y = 3e^x \cos 2x + 2xe^{3x}$ 5 M L3
- 8 a. State and prove the relation between Beta and Gamma Function. 6 M L4
- b. Evaluate $\int_0^2 x^3 \sqrt{2-x} dx$ using Beta And Gamma function. 4 M L5
- OR
- c. State and Prove Rodrigue's Formula. 6 M L4
- d. Evaluate $\int_0^a x^4 \sqrt{a^2 - x^2} dx$ 4 M L5

- 9 a. Verify Cayley – Hamilton Theorem for the Matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$ and also find A^{-1} 6 M L4
- b. Discuss the nature of the Q.F. $x^2 + 4xy + 6xz + 2yz - y^2 + 4z^2$ 4 M L4
- OR**
- c. Find Eigen Values and corresponding Eigen Vectors of $\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ 8 M L3
- d. Using Cayley – Hamilton Theorem, find A^8 if $A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$ 2 M L3
- 10 a. Find the Laplace Transform of (i) $t e^{2t} \sin 3t$ (ii) $e^{-t} \cos^2 t$ 5 M L3
- b. Find the Inverse Laplace Transform of $\frac{2s+12}{s^2+6s+12}$ and $\frac{s}{(s+2)^2}$ 5 M L3
- OR**
- c. Find $L \{ t \cos at \}$ 2 M L5
- d. Using Laplace Transform method, solve $(D^2 + 1) y = 6 \cos 2t$, $t > 0$, If $y = 1$, $Dy = 1$ when $t = 0$. 8 M L5

Stanley College of Engineering and Technology for Women (A)

B.E (CSE/EEE/ECE/IT/CME/AI&DS) I Semester (Backlog) Examinations January-2023

Mathematics-I

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

1. Test the convergence of the series $\sum (1 + \frac{1}{n})^{-n^2}$ 2M
2. Verify Cauchy's mean value theorem for $f(x) = x^2$ and $g(x) = x^3$ in $[1, 2]$ 2M
3. If $u = \frac{x+y}{1-xy}$ and $v = \tan^{-1} x + \tan^{-1} y$ then prove that u and v are functionally dependent. 2M
4. Find grad f if $f = x^2y + y^2x + z^2$ at $(1, 1, -2)$ 2M
5. Find the first approximation of the root lying between 0 and 1 of the equation $x^3 + 3x - 1 = 0$ by Newton-Raphson formula. 2M

PART B

Note: Answer all questions

5 X 10 = 50M

6. (a) Discuss the convergence or divergence of $\frac{x}{1.2} + \frac{x^2}{2.3} + \frac{x^3}{3.4} + \dots$ 5M
- (b) Examine the convergence of the series $1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \dots$ 5M

(OR)

- (c) Test the convergence of the series $1 + \frac{2}{5}x + \frac{6}{9}x^2 + \dots + \frac{2^n-2}{2^{n+1}}x^{n-1} + \dots$ ($x > 0$) 5M
- (d) Show that the exponential series $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ converges absolutely for all values of x . 5M
7. (a) Find the evolute of the parabola $y^2 = 4ax$ 5M
- (b) Verify Cauchy's Mean value theorem for the functions $f(x)$ and $f^1(x)$ in $(1, e)$ given $f(x) = \log x$ 5M

(c) Prove that $\frac{\pi}{6} + \frac{1}{5\sqrt{3}} < \sin^{-1}\left(\frac{3}{5}\right) < \frac{\pi}{6} + \frac{1}{8}$ 10M

8. (a) If $u = x + y + z$, $y + z = uv$, $z = uvw$ then show that $\frac{\partial(x,y,z)}{\partial(u,v,w)} = u^2v$ 5M

(b) If $u = x^2 + y^2$, $x = at^2$, $y = 2at$ then find $\frac{du}{dt}$ 5M

(OR)

(c) A rectangular box open at the top is to have volume 32 cubic ft. Find the dimensions of the box requiring least material for its construction. 5M

(d) Find the maximum and minimum values of $f(x,y) = x^3 + y^3 - 3axy$ 5M

9. (a) If $F = (5xy - 6x^2)i + (2y - 4x)j$ Evaluate $\int_c \vec{F} \cdot d\vec{r}$ where c is the curve in the xy plane $y = x^3$ from $(1,1)$ to $(2,8)$ 5M

(b) Find $\text{div } \vec{f}$ when $\vec{f} = \text{grad}(x^3 + y^3 + z^3 - 3xyz)$ 5M

(OR)

(c) Verify Green's theorem in the plane for $\oint (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is the region bounded by $y = \sqrt{x}$ and $y = x^2$ 10M

10. (a) Solve the following system of equations using Gauss Seidel iteration method

$10x + 2y + z = 9$, $x + 10y - z = -22$, $-2x + 3y + 10z = 22$ 10M

(OR)

(c) Using Newton's divided difference formula evaluate $f(8)$ given that 10M

x	4	5	7	10	11	13
f(x)	48	100	294	900	1210	2028

Hall Ticket No-

Code: S222822/1

Stanley College of Engineering and Technology for Women(A)

B.E (IT) I Semester (Backlog) Examinations January-2023

Chemistry

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- | | | |
|---|--|------------|
| 1 | Differentiate between anodic and cathodic coatings. | 2 M CO1L2 |
| 2 | Give the use of photovoltaic cells in harnessing the solar energy. | 2 M CO2 L1 |
| 3 | Outline the preparation of Kevlar, mention any two properties. | 2 M CO3 L4 |
| 4 | Define HCV and LCV. | 2 M CO4 L1 |
| 5 | Explain any two examples of clean technology. | 2 M CO5 L2 |

PART-B

Note: Answer all questions

5 X 10= 50M

- | | | | |
|---|----|---|------------|
| 6 | a. | Explain the principle of estimation of hardness of water by complexometric titration. | 6 M CO1 L2 |
| | b. | 100 ml of a standard hard water containing 0.2 mg/ml of pure CaCO_3 consumes 25 ml of EDTA. 100 ml of test water sample consumes 35 ml of same EDTA before boiling and 20 ml after boiling. Calculate the temporary, permanent and total hardness of water sample. | 4 M CO1 L3 |
| | | OR | |
| | c. | With a neatly labeled diagram, explain demineralization of water by ion exchange method. Enlist the advantages and limitations. | 6 M CO1 L2 |
| | d. | Discuss about galvanizing process. | 4 M CO1 L2 |
| 7 | a. | Illustrate the construction and working of quinhydrone electrode. | 5 M CO2 L4 |
| | b. | Demonstrate the determination of pH of a solution by using this electrode with a diagram. | 5 M CO2 L3 |
| | | OR | |
| | c. | Illustrate the electrochemistry of Lead acid battery. Write its merits and applications. | 5 M CO2 L4 |
| | d. | Define fuel cell. Explain the construction and working of methanol-oxygen fuel cell. | 5 M CO2 L1 |

Code: S222822/1

- a. Outline the synthesis of silicone rubber. Mention its properties and applications. 5 M CO3 L4
- b. Illustrate the molecular orbital energy level diagram of O₂. 5 M CO3 L4

OR

- c. What is PET? How is it prepared from its monomers? Write its properties and uses. 5 M CO3 L1
- d. Explain the importance of biodegradable polymers with an appropriate example along with its preparation, properties and applications. 5 M CO3 L2

- a. Calculate the gross, net calorific value and the volume of air at STP, required for the complete combustion of 2 Kg of the fuel having the following composition. 78% carbon, 8 % hydrogen, 2.1 % sulfur, 1.6% nitrogen and 4% ash. 5 M CC L3
- b. What is the necessity and significance of elemental analysis of coal? How can you analyze coal with the help of ultimate analysis? 5M CO4 L1,4

OR

- c. Classify fuels and give appropriate examples. Enumerate the requisites of a good fuel. Write the composition and uses of LPG, CNG. 5M CO4 L1,6,4
- d. Explain about knocking. How do you rate the quality of petrol and diesel? 5 M CO4 L2

- a. Composite materials have superior properties compared to conventional materials. Substantiate. 5 M CO5 L5

- b. Give an overview of trans esterification. Enlist the applications of biodiesel. 5 M CO5 L1

OR

- c. Enumerate the applications of nano materials. 4 M CO5 L2
- d. What is green chemistry? Explain the principles of green chemistry. 6M CO5 L1,2

Hall Ticket No-

Code: S22325

Stanley College of Engineering and Technology for Women(A)

B.E (ECE,EEE & IT) I Semester Backlog Examinations-January-2023

Environmental Science

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- | | | | |
|---|--|----|----|
| 1 | List the effects of deforestation on tribal people. | 2M | L1 |
| 2 | Discuss the two channel energy flow in an ecosystem. | 2M | L5 |
| 3 | List the various bio geographical zones in India. | 2M | L1 |
| 4 | List the salient features of air act. | 2M | L1 |
| 5 | Define global warming. | 2M | L1 |

PART-B

Note: Answer all questions

5 X 10= 50M

- | | | | |
|----|--|----|----|
| 6 | a. Discuss the various ill-effects of modern agriculture. | 5M | L5 |
| | b. Relate soil erosion impact desertification and fertility. | 5M | L1 |
| | OR | | |
| | c. Discuss soil erosion and desertification. | 5M | L5 |
| | d. Explain problems of Dam. | 5M | L3 |
| 7 | a. Discuss the function of an ecosystem. | 5M | L5 |
| | b. Explain about desert ecosystem. | 5M | L3 |
| | OR | | |
| | c. Analyze the structure of Ecosystem. | 5M | L4 |
| | d. Explain about Types of Ecological pyramids. | 5M | L4 |
| 8 | a. Explain about endangered and endemic species of India. | 5M | L3 |
| | b. Describe the conservation of biodiversity. | 5M | L5 |
| | OR | | |
| | c. Write the major threats to biodiversity. | 5M | L1 |
| | d. Relate the Values of Biodiversity. | 5M | L1 |
| 9 | a. Explain the causes of water pollution. | 5M | L1 |
| | b. Write about Wildlife Act and Forest conservation Act. | 5M | L1 |
| | OR | | |
| | c. Explain about Environmental Legislation. | 5M | L1 |
| | d. Discuss effects and control measures of Air pollution. | 5M | L5 |
| 10 | a. Analyze mitigation measures for Ozone layer depletion. | 5M | L4 |
| | b. Explain about disaster management of India. | 5M | L1 |
| | OR | | |
| | c. Explain about Acid rains. | 5M | L2 |
| | d. Categorize Types of Disasters. | 5M | L3 |